

# TOPICS

*to be covered*

**1** Type topic here

**2** Type topic here

**3** Type topic here

**4** Type topic here

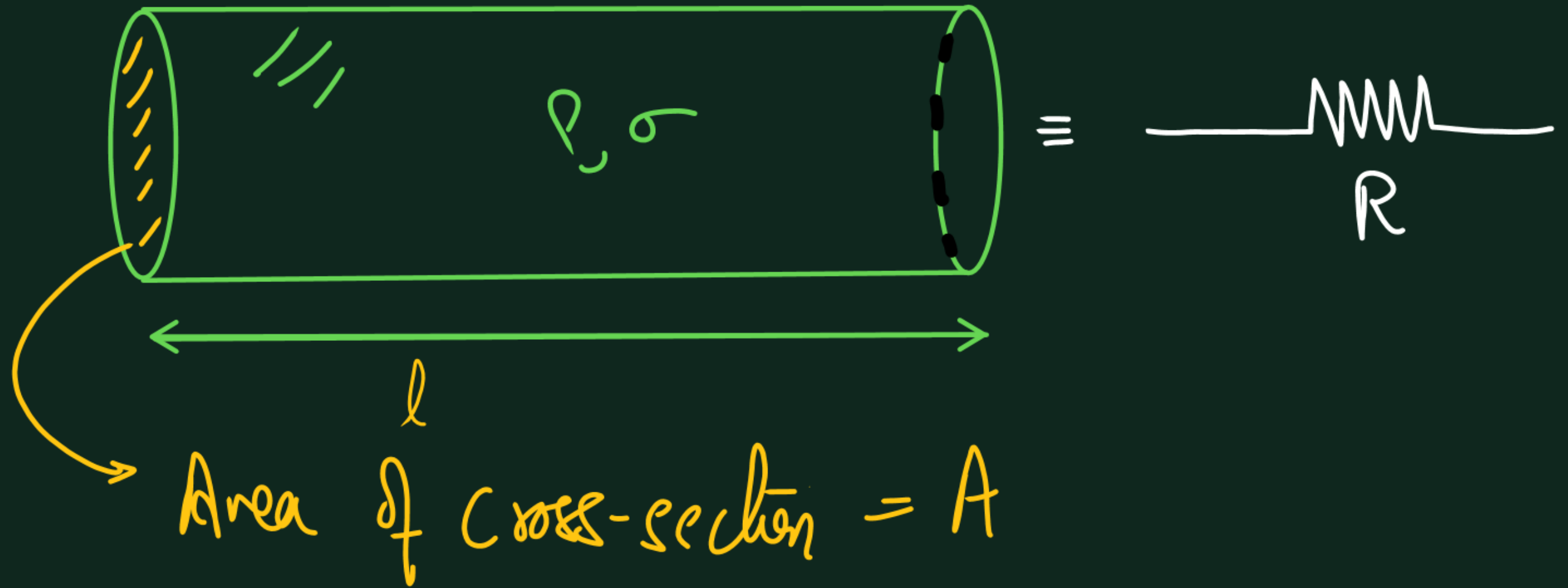
Circuit Elements R

Voltage & Current Source

Power Calculation



Resistance  $R$  :- of a wire length =  $l$



$$R = \frac{l}{\sigma A} = \rho \frac{l}{A}$$

$$\rho = \frac{1}{\sigma}$$

$\sigma \rightarrow$  Conductivity of Material

$\rho \rightarrow$  Resistivity of Material

$\hookrightarrow$  or specific Resistance

For Conductors eg Al, Cu etc

$$\rho = \frac{1}{\sigma}$$



$\sigma \rightarrow$  high if  $\rho \rightarrow$  low

$\Rightarrow R \rightarrow$  small

For Insulators eg, Plastic, wood etc

$\sigma \rightarrow$  low,  $\rho \rightarrow$  High

$R \rightarrow$  High

DC battery or cell or Voltage source. -  
 ↓  
 Direct Current

$E \rightarrow$  EMF of  
 battery or  
 source



or

or



DC battery or cell or Voltage source:-

↓  
Direct Current

EMF → Electro-Motive Force  
↓

$E \rightarrow$  EMF of  
battery or  
source

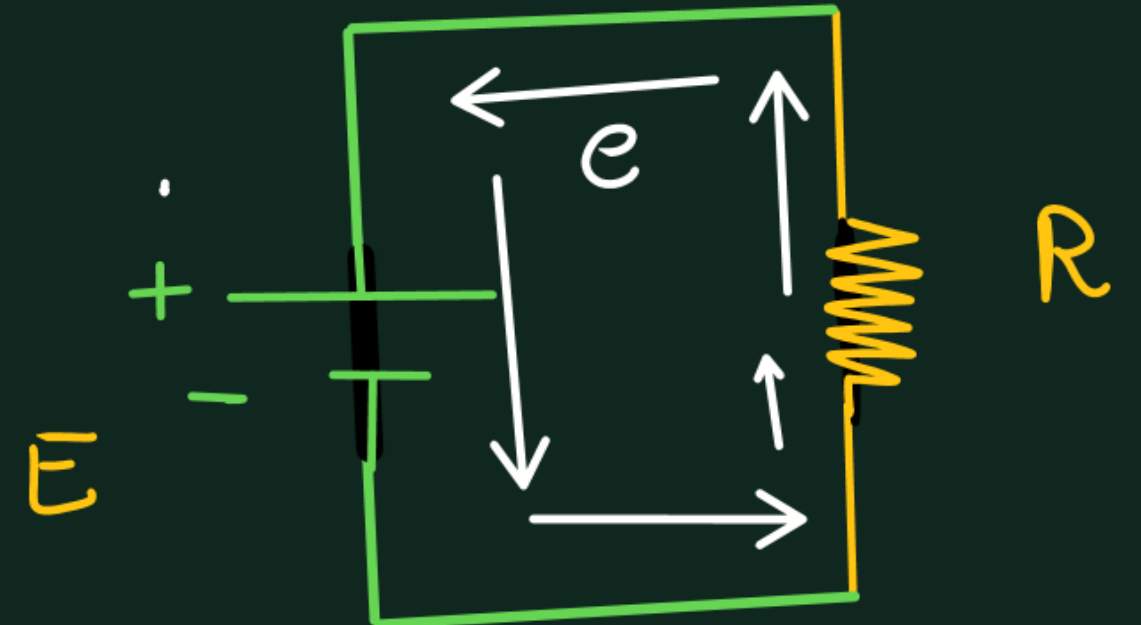
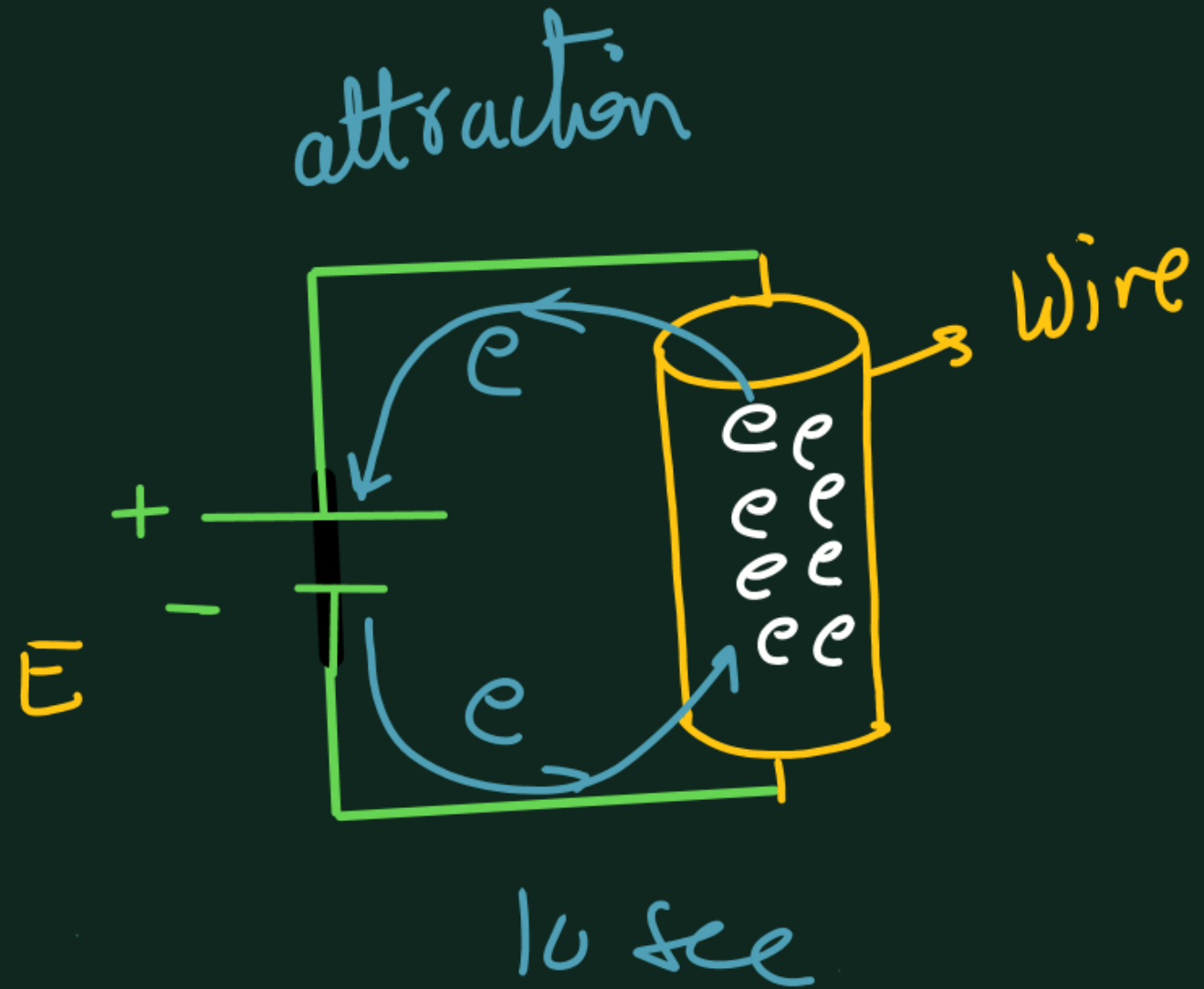
eg:- 12V battery

V-Volt

1.5V

5V

$$e = -1.6 \times 10^{-19} \text{ C} \rightarrow \text{Coulombs}$$



Electric Current :- Rate of flow of charge.

i.e. charge flowing per sec

eg:-  $n = 5 \times 10^{20}$  electrons are flowing in 10 sec

In time  $t = 10$  sec, charge flowing

$$q = n \cdot e = 5 \times 10^{20} \times 1.6 \times 10^{-19} \\ = 80 \text{ coulomb}$$

Current  $I = \frac{q}{t} = \frac{80 \text{ C}}{10 \text{ s}} = 8 \text{ C/s} = 8 \text{ A}$

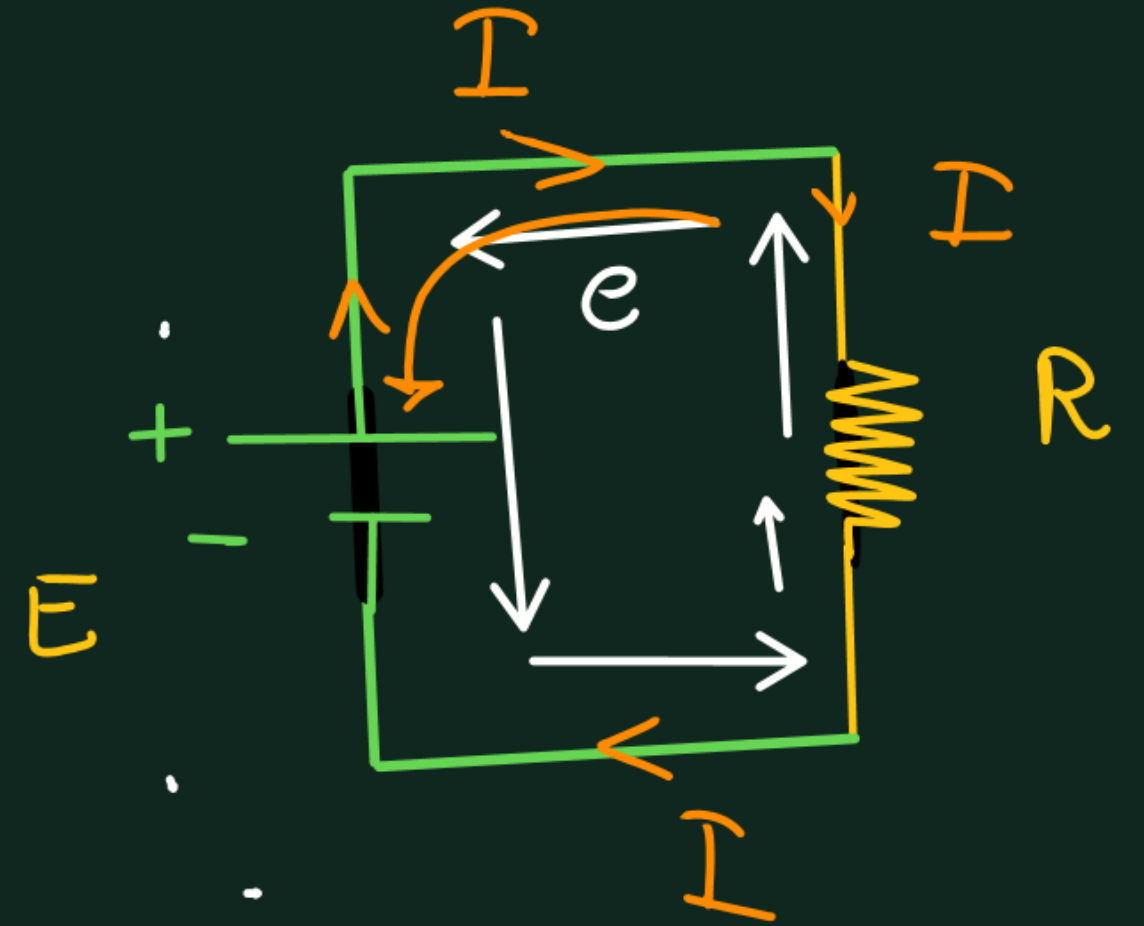
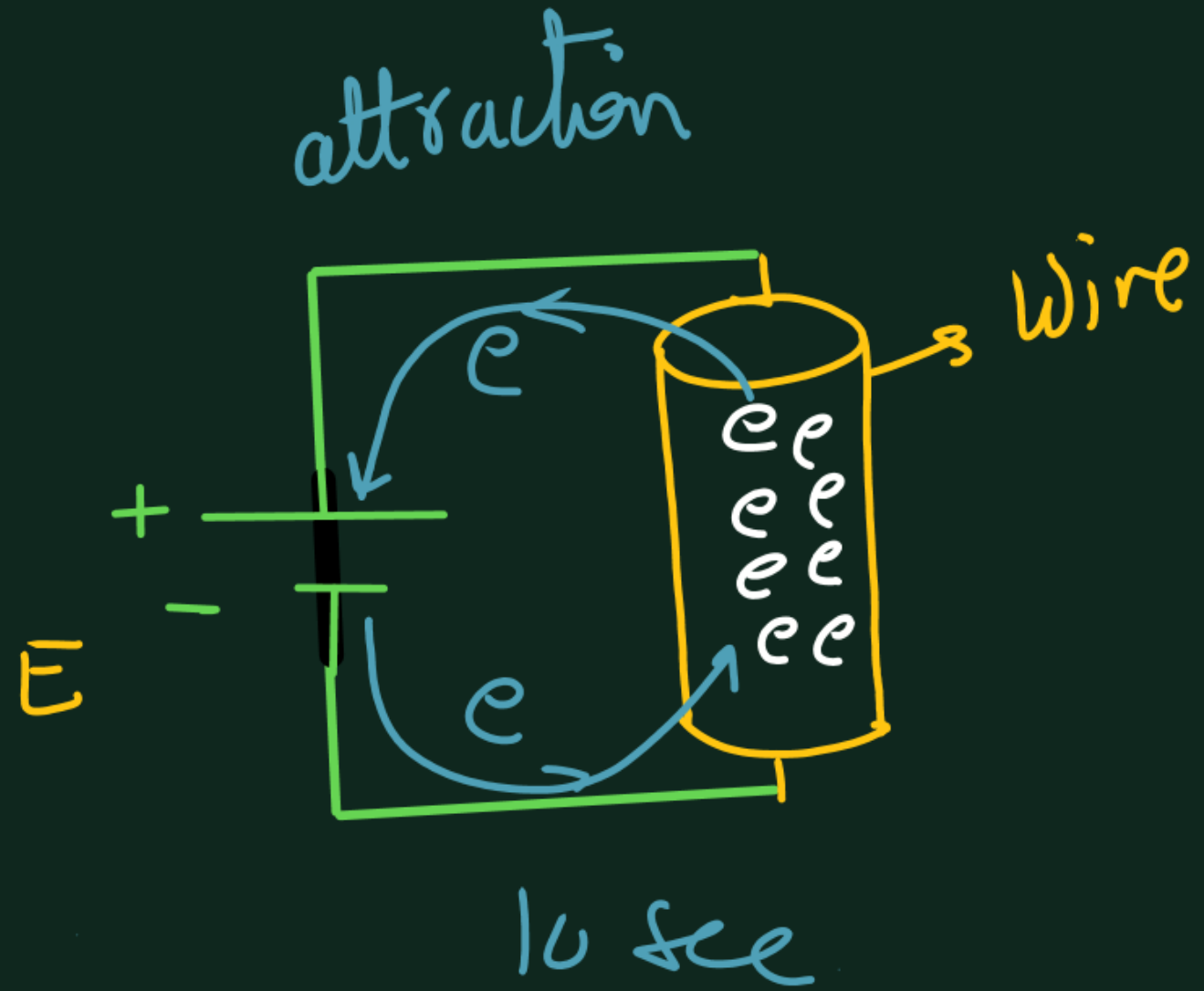


Coulomb/sec = Ampere

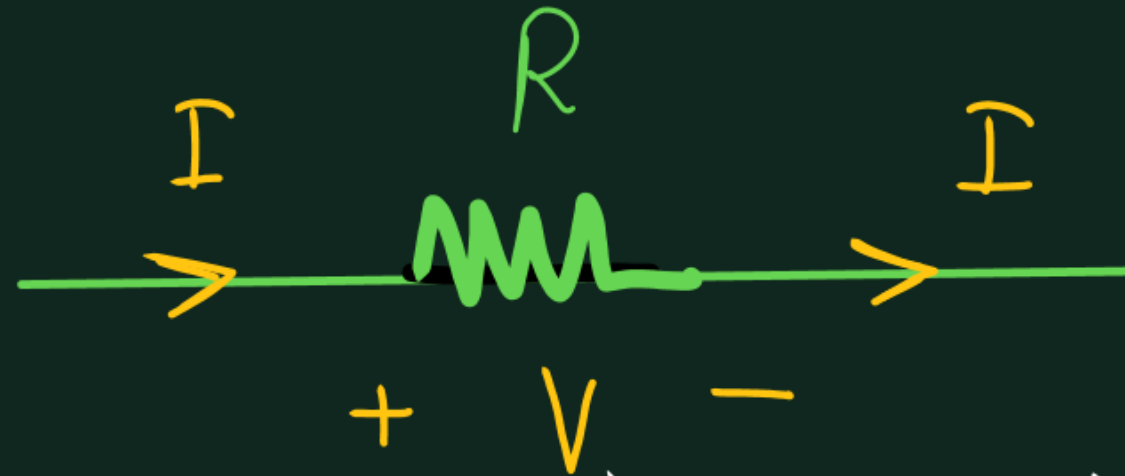
Standard Convention:-

direction of Current  $\longrightarrow$   
In the direction of +ve charge  
or In the opposite direction of -ve charge

$$e = -1.6 \times 10^{-19} \text{ C} \rightarrow \text{Coulombs}$$



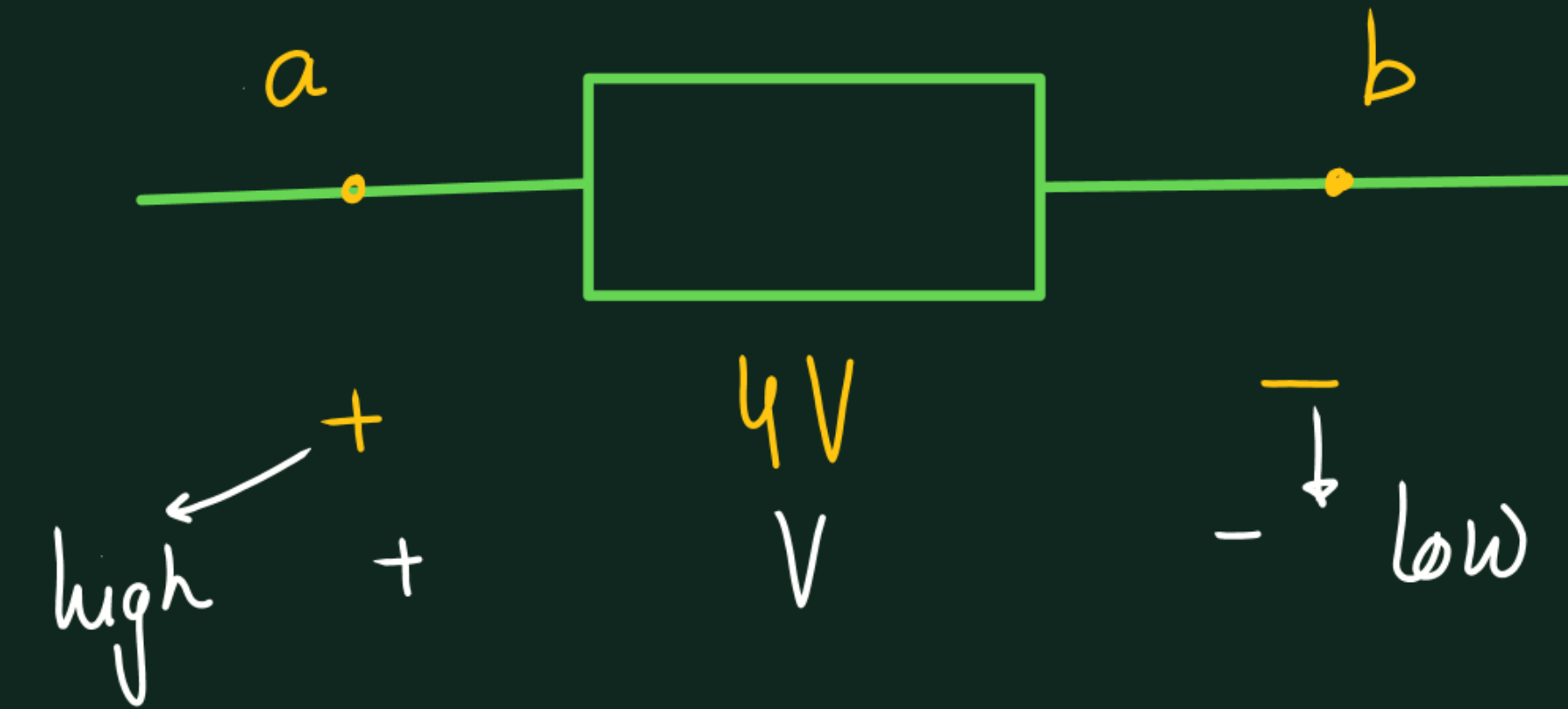
ohm's law :-



Voltage drop across resistor

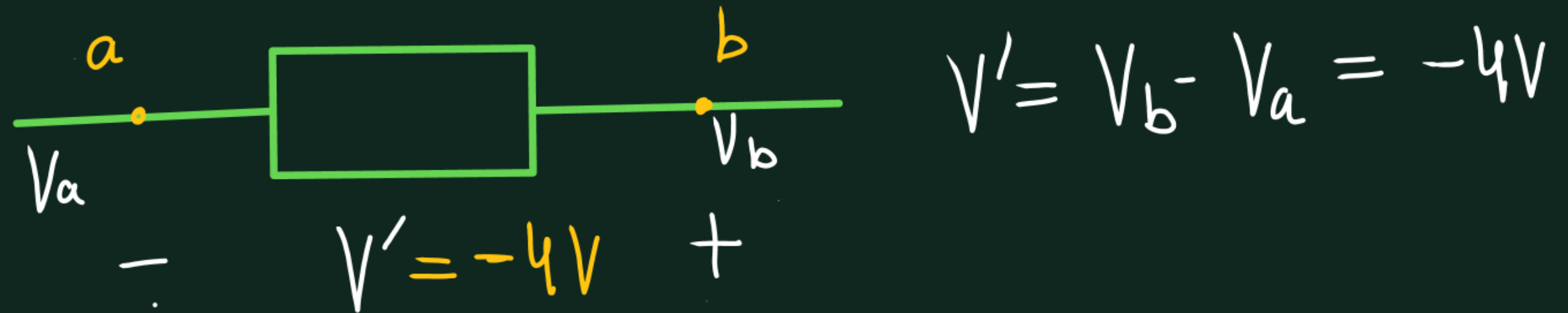
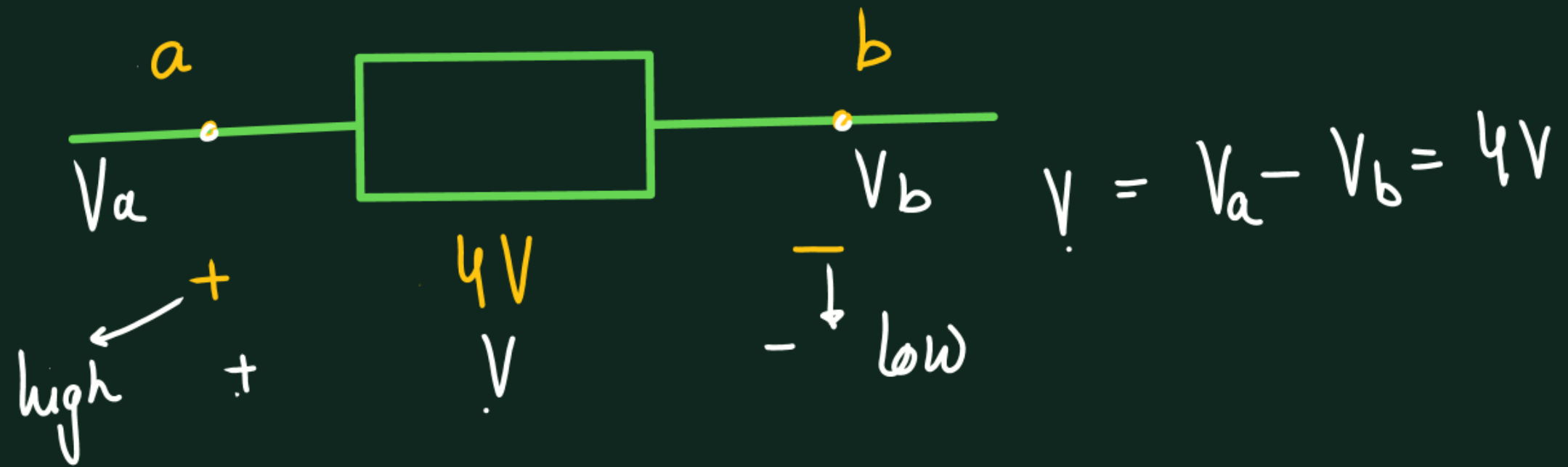
$$V = IR \Rightarrow I = V/R$$

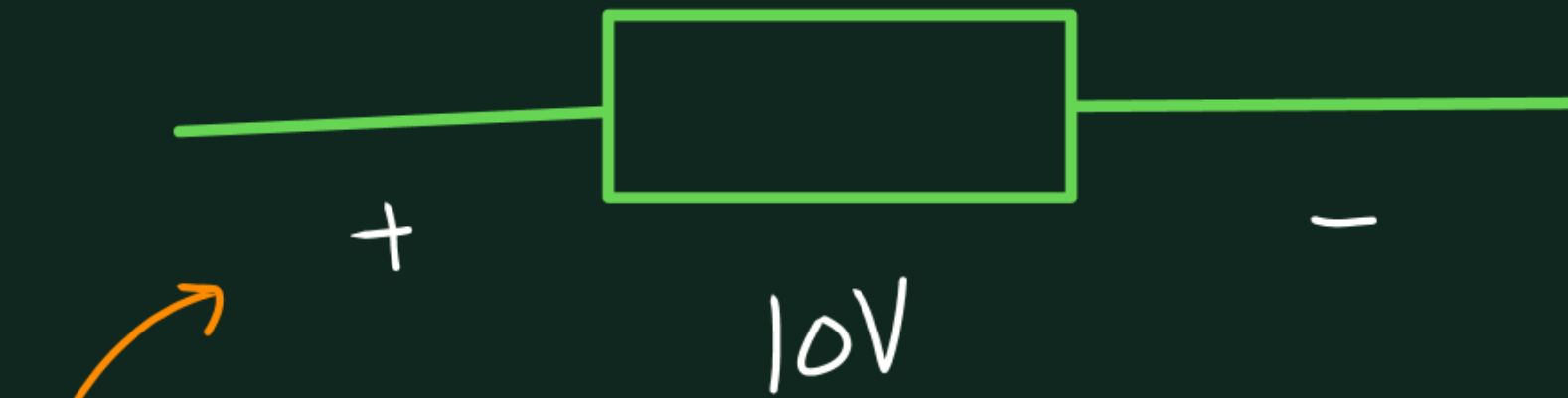
$$R = \frac{V}{I} \rightarrow \frac{\text{Volt}}{\text{Ampere}} = \Omega \text{ ohm}$$



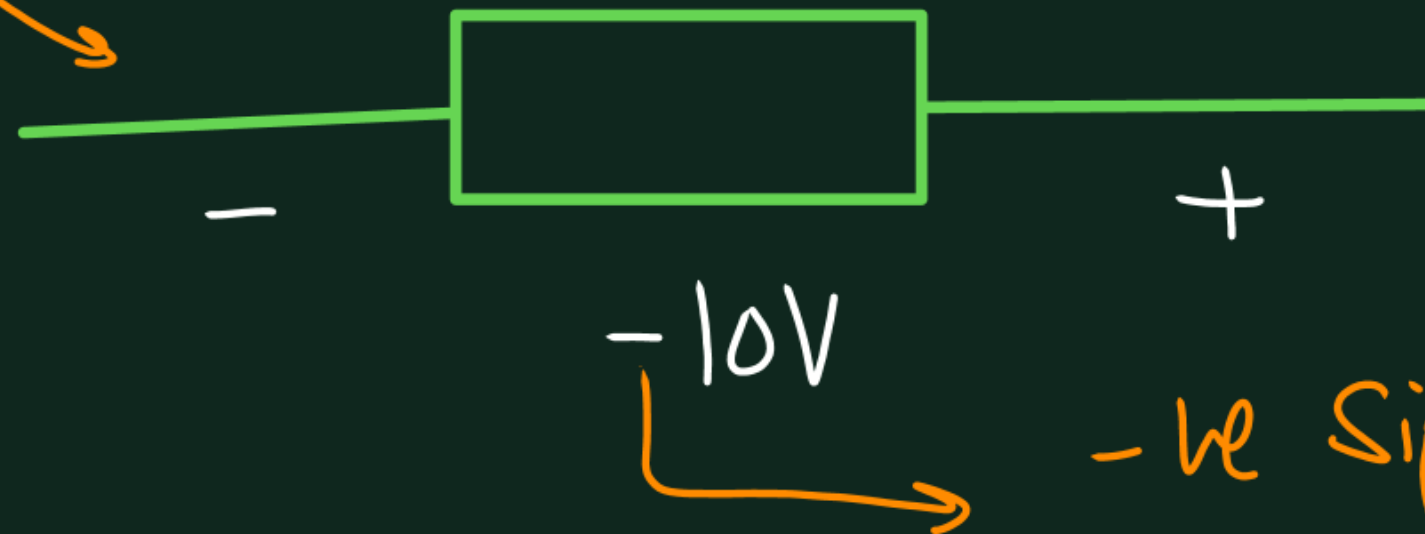
Potential  
or  
Voltage  
difference

$$V = V_a - V_b = 4V$$

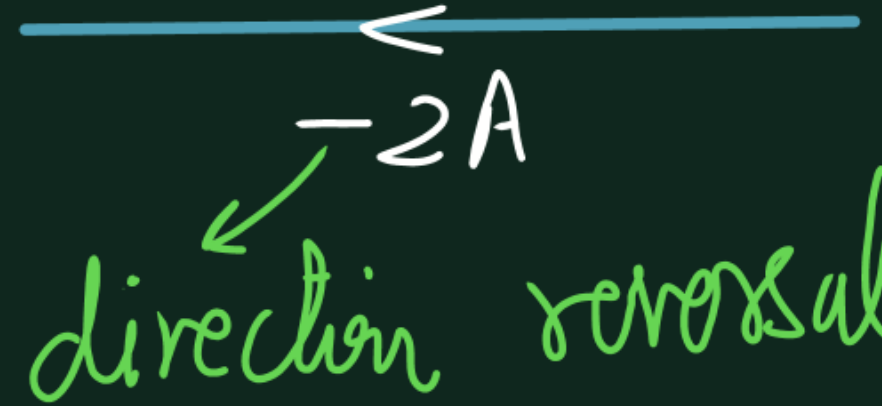
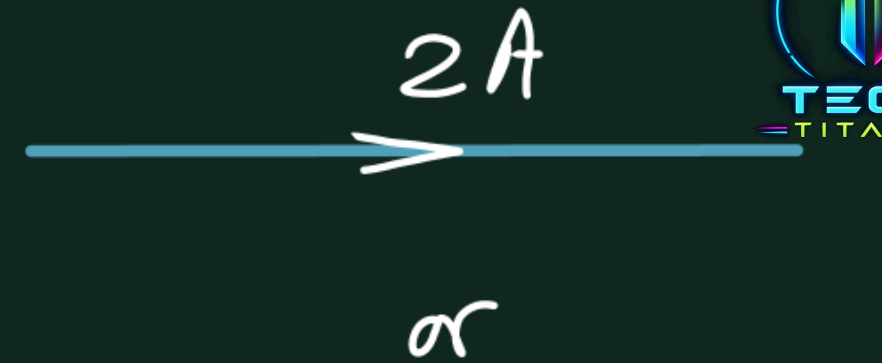




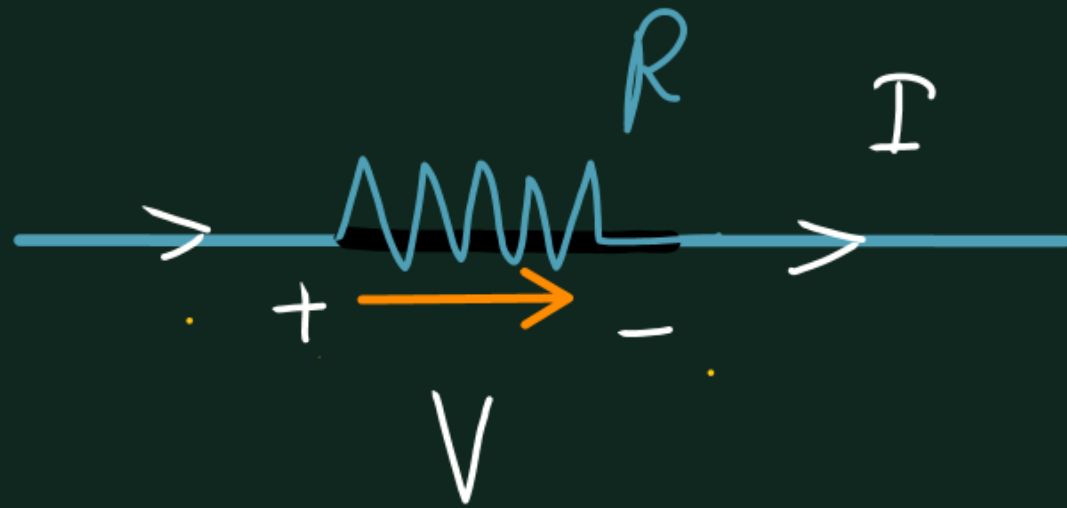
OR



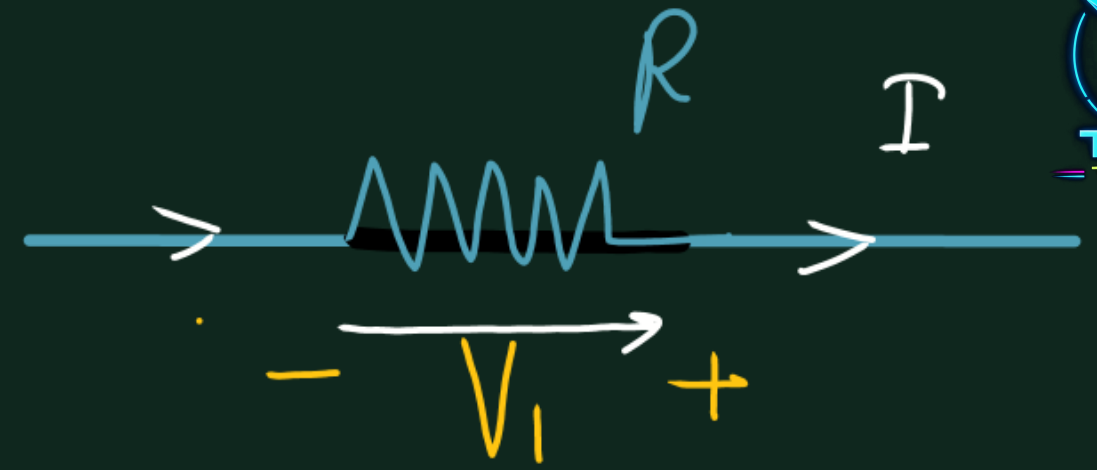
-ve sign  $\Rightarrow$  Polarity reversal



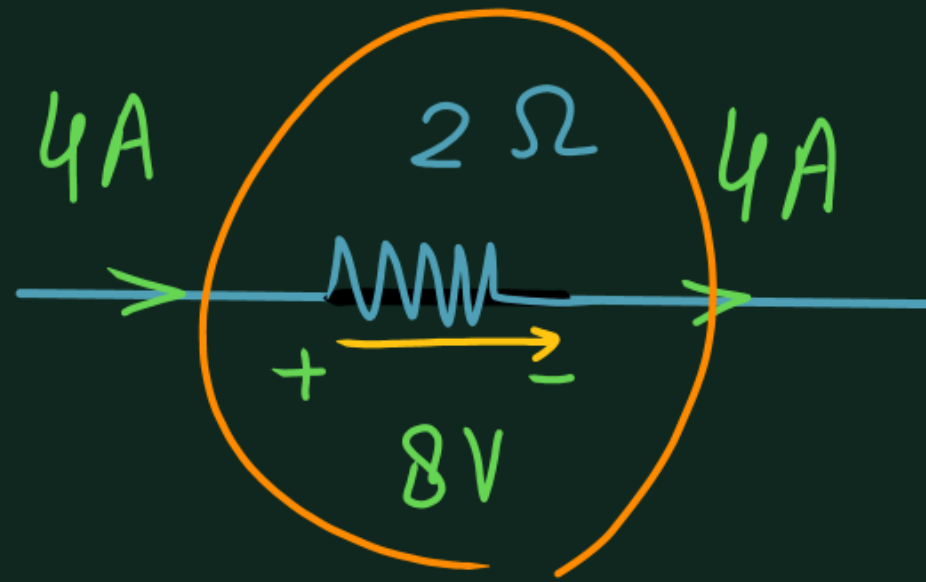
direction reversal



$$V = IR$$



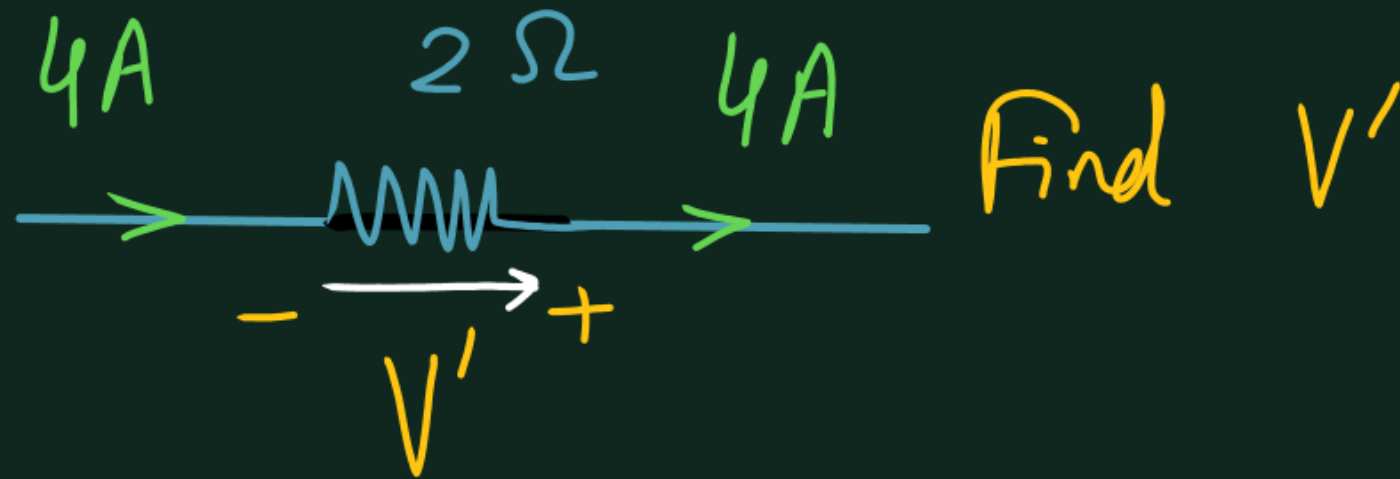
$$V_1 = -IR$$



$$R = 2\ \Omega, \quad I = 4A$$

Voltage

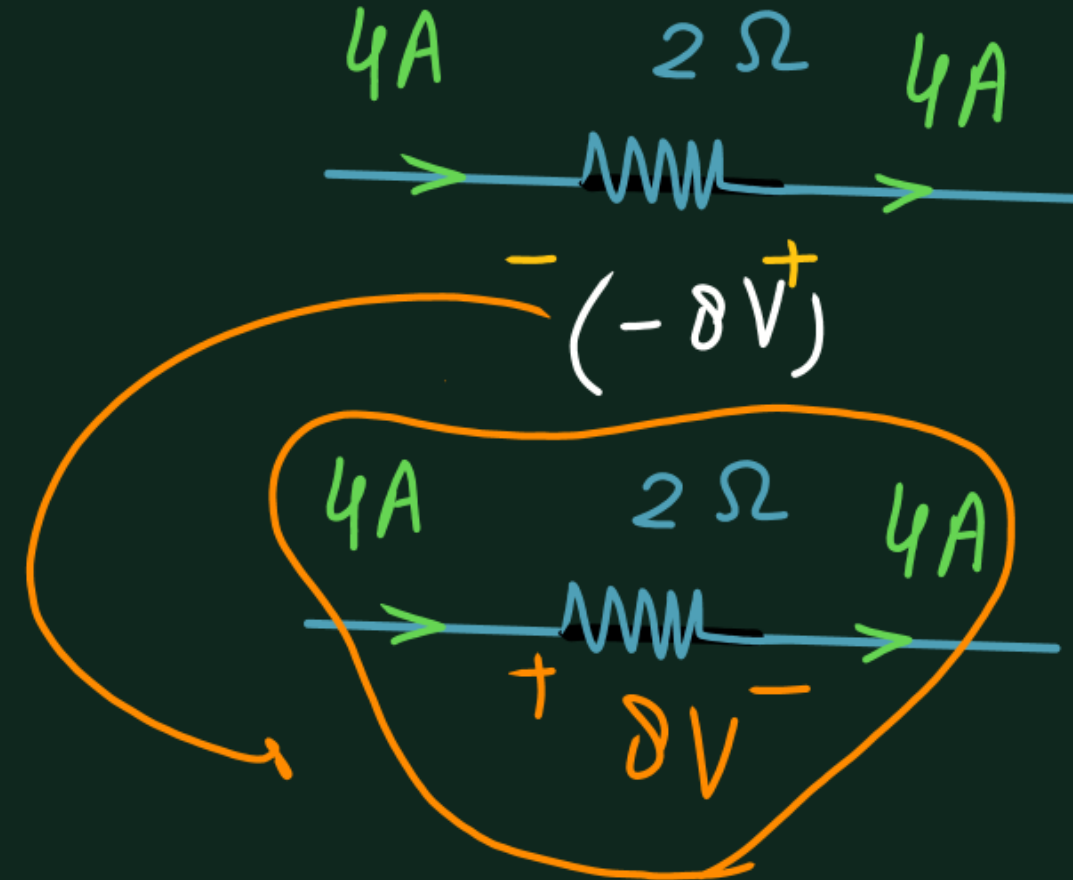
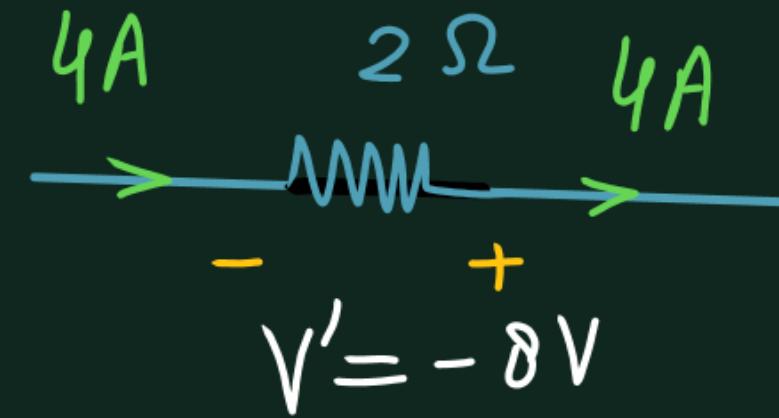
$$V = IR = 2 * 4 = 8V$$



$$R = 2\Omega, \quad I = 4A$$

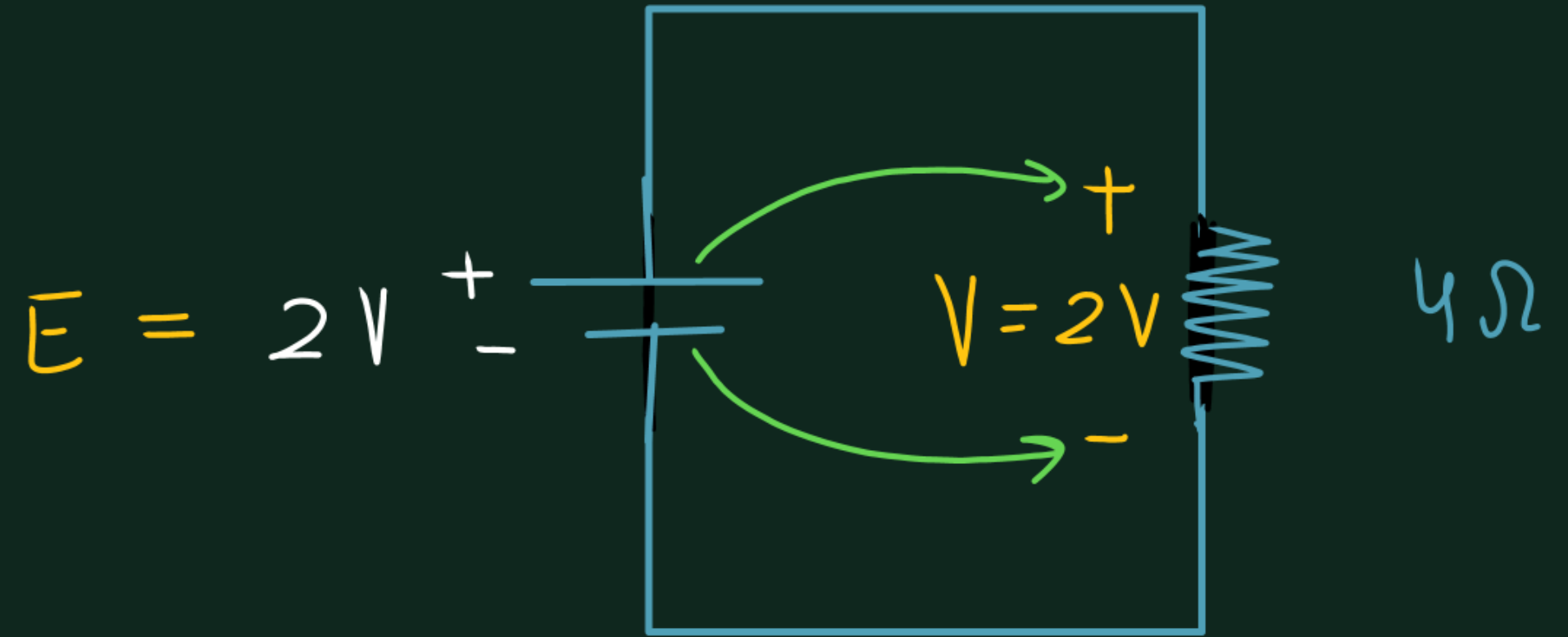
$$V' = -IR = -2 \times 4$$

$$= -8V$$



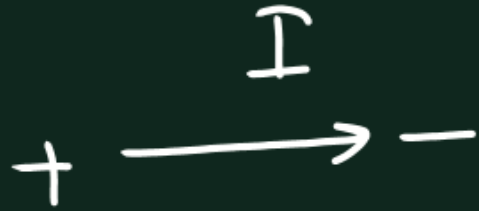
Ques-

$$I = \frac{V}{R}$$



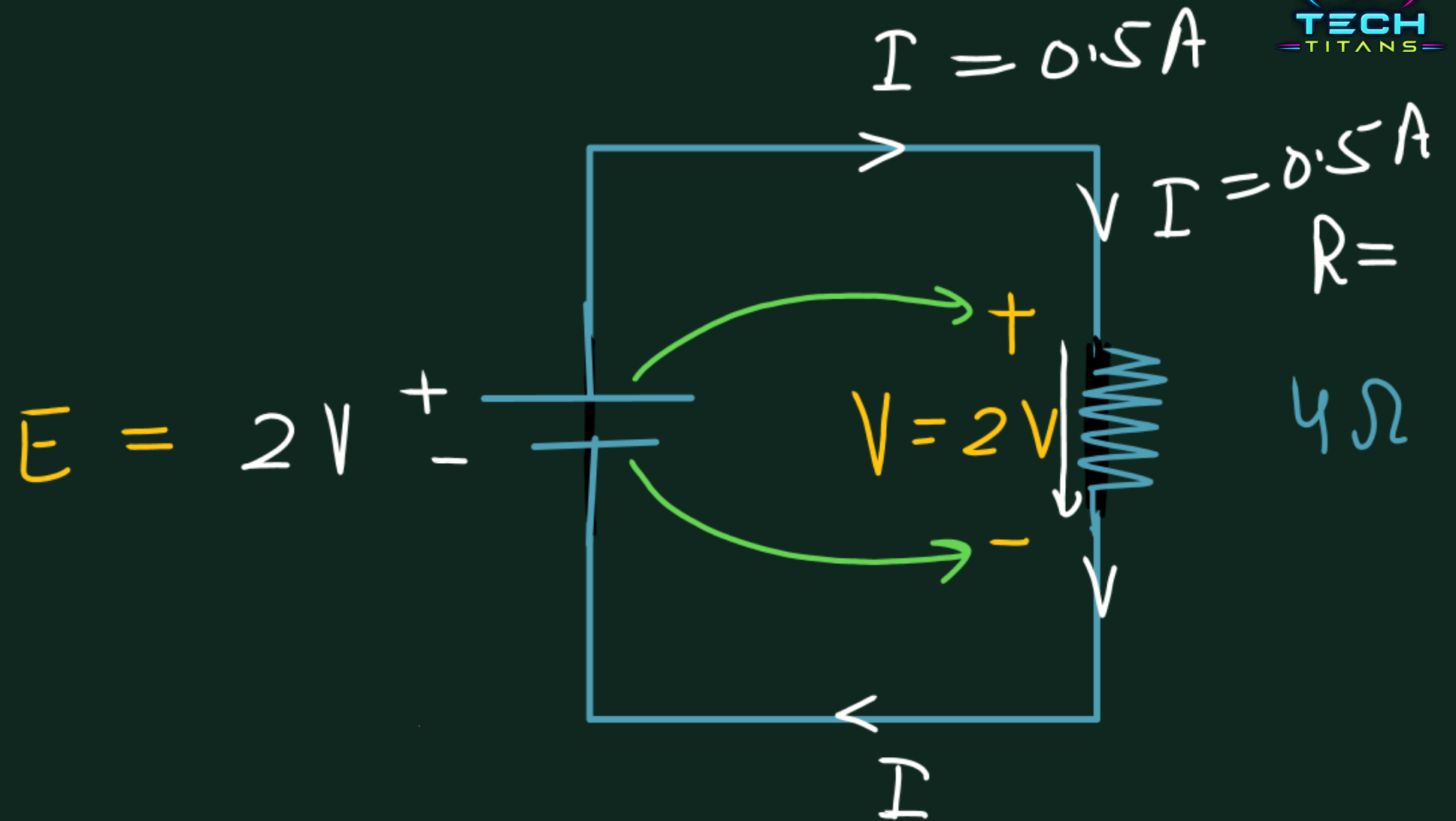
Ques-

$$I = \frac{V}{R}$$



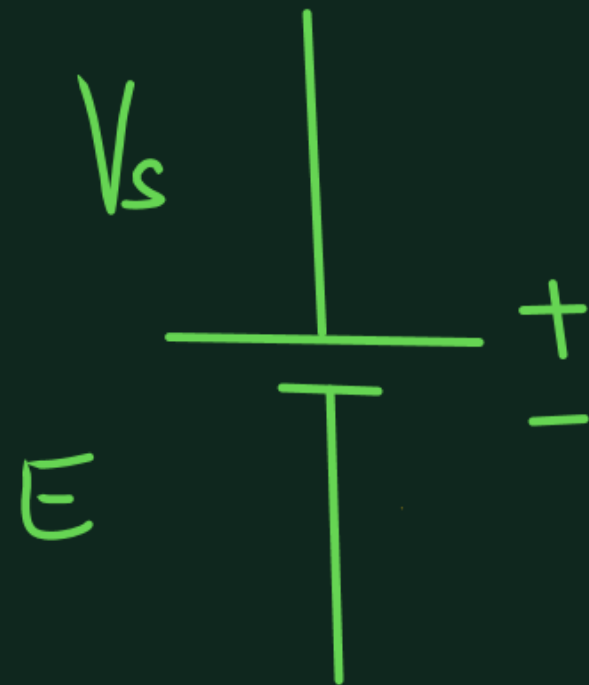
$$I = \frac{V}{R} = \frac{2}{4} = 0.5 \text{ A}$$

$$E = 2 \text{ V}$$

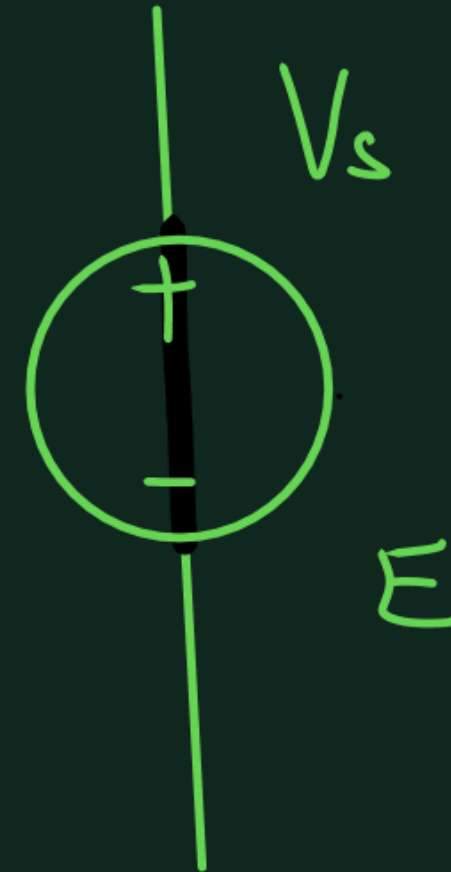


# Voltage Source & Current Source

## 1. Voltage Source:- $E \propto V_s$



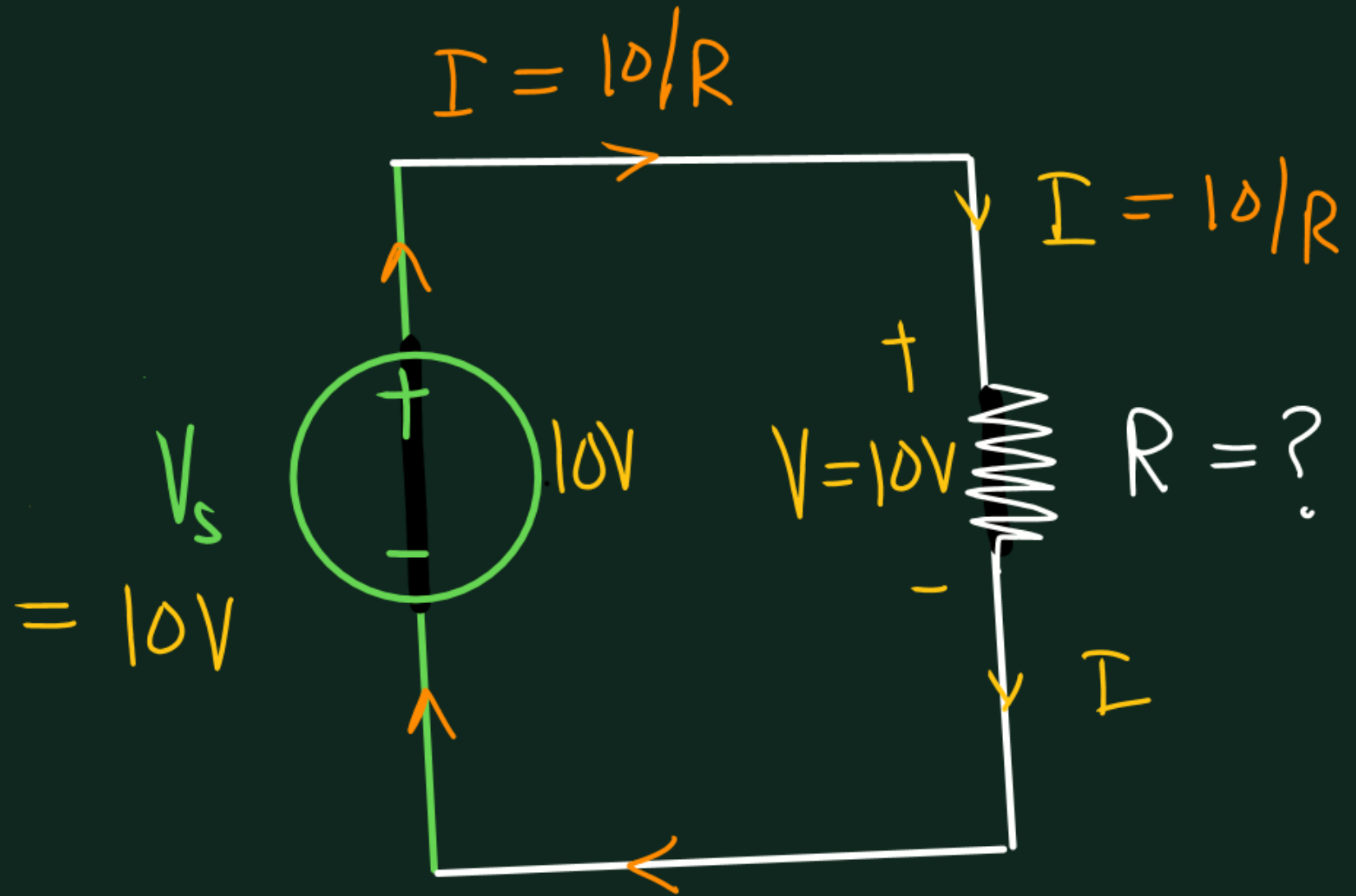
or



Voltage source:- It maintains constant voltage across its terminals whatever be the current flowing



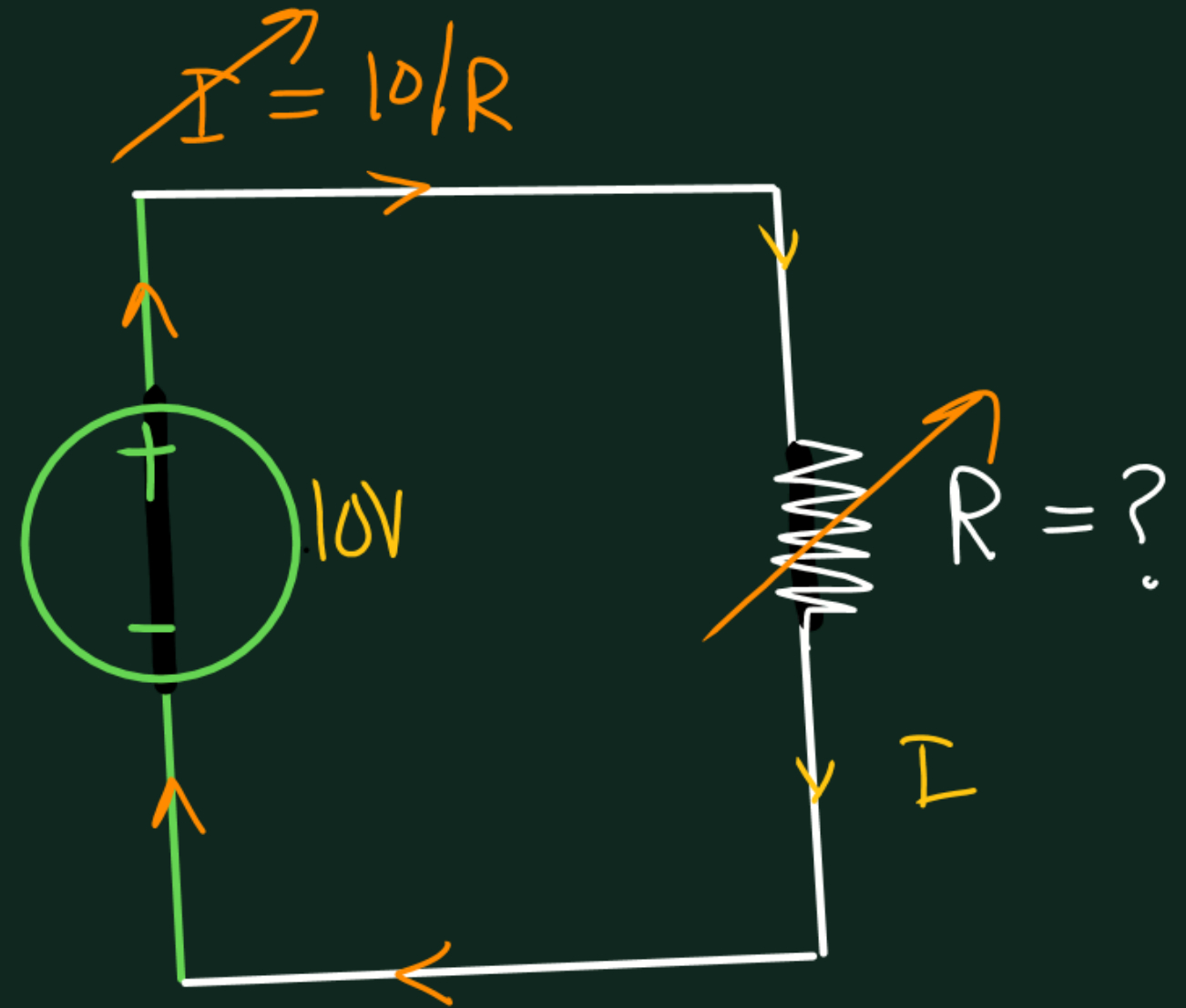
$$I = \frac{V_s}{R} \Rightarrow I = \frac{10}{R}$$



Voltage source:- It maintains constant voltage across its terminals whatever be the current flowing



$$I = \frac{V_s}{R} \Rightarrow I = \frac{10}{R}$$



2. Current Source:- It provides constant current



whatever be the voltage.

$$I_s = 4A$$

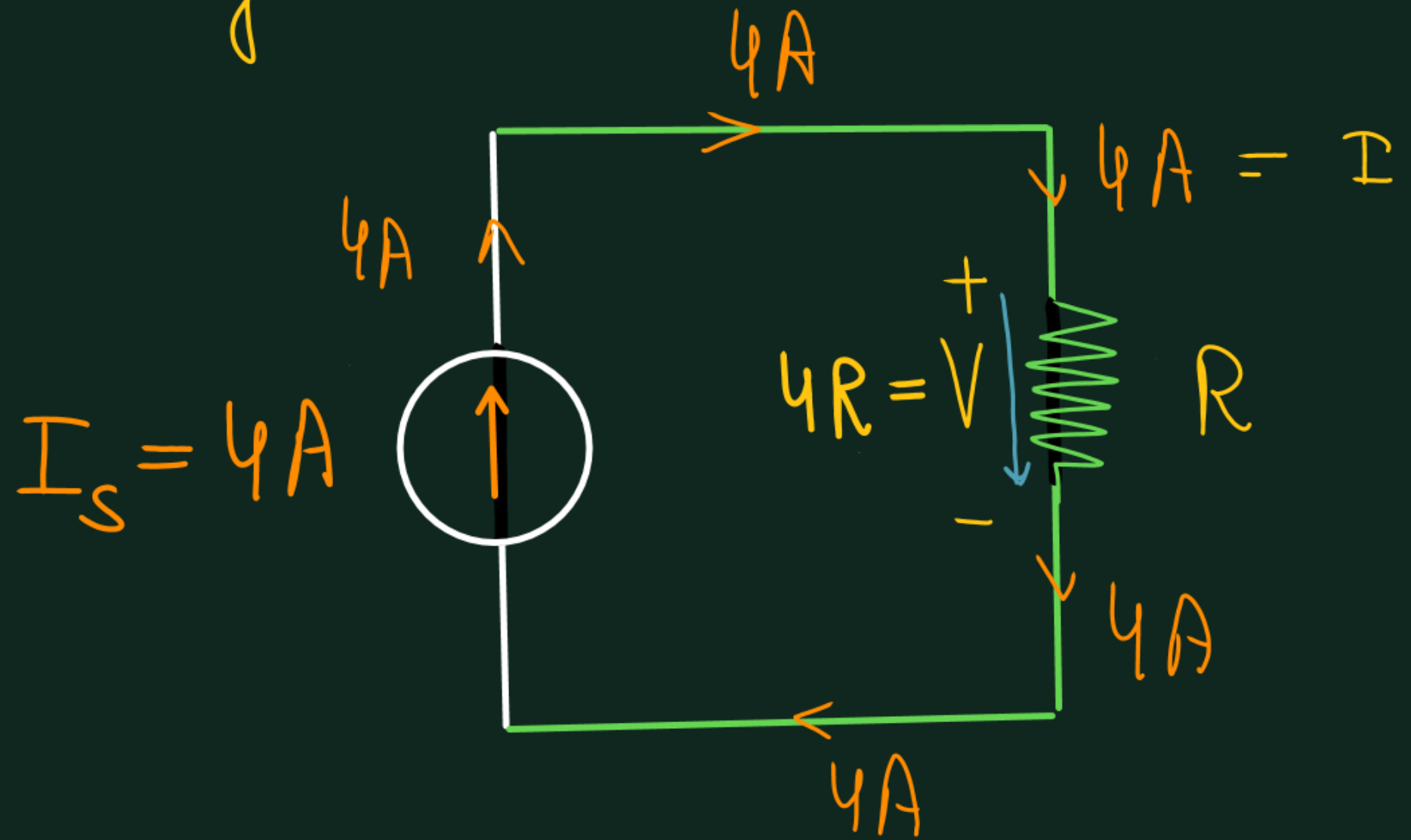


2. Current Source:- It provides constant current

whatever be the voltage.

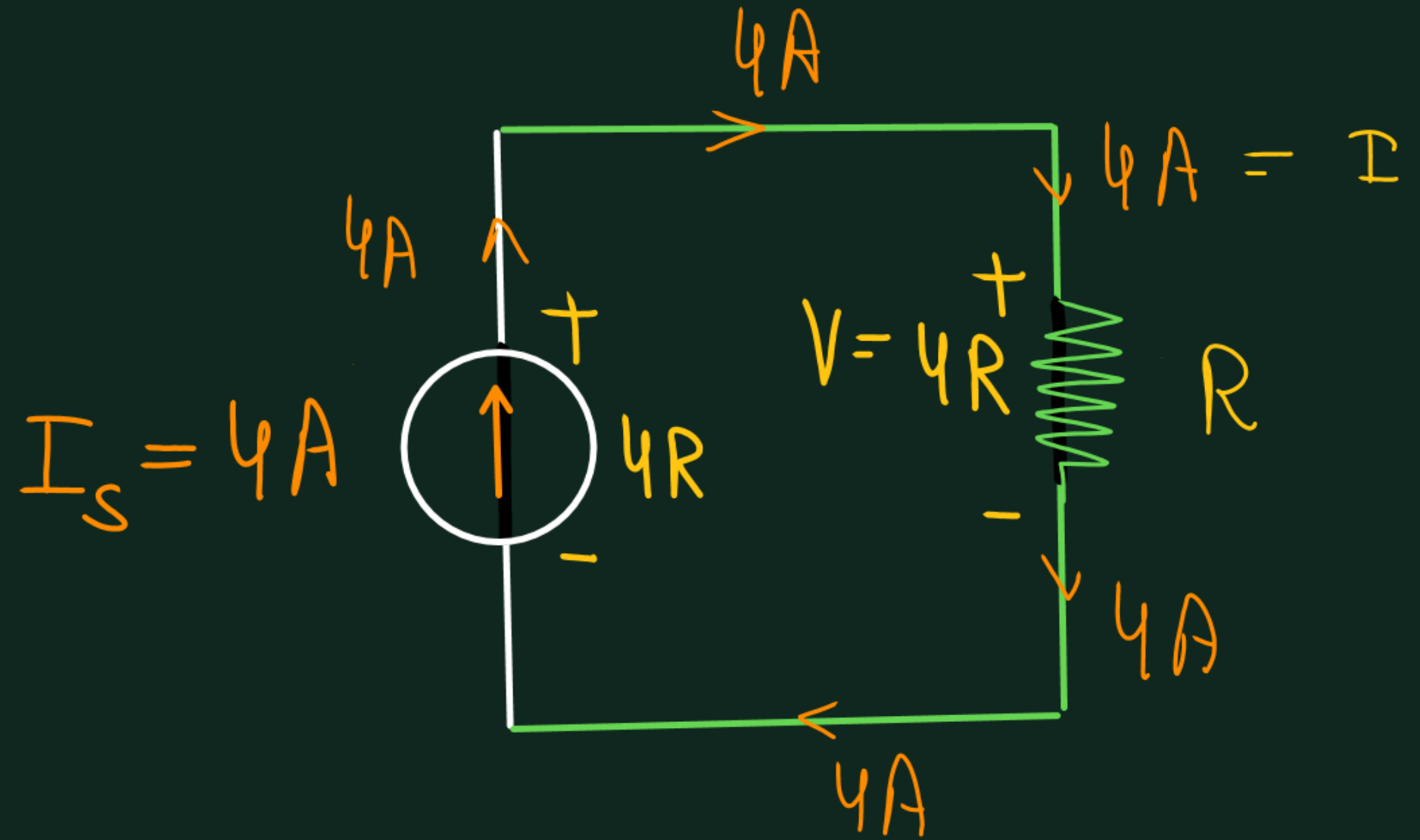
$$V = IR$$

$$V = 4R$$



$$V = IR$$

~~$V = 4R$~~



$$V = IR$$

$$\cancel{V = 4R}$$

